

Executive Summary

Problem

Emergency radiology departments process a continuous stream of CT brain studies with varying levels of clinical urgency. Although every scan requires expert review, patients with acute intracranial hemorrhage (ICH) require immediate diagnosis and treatment to minimize neurological damage and improve survival. In current workflows, CT studies are typically reviewed in chronological order or based on manually assigned priorities, which may delay the identification of life-threatening hemorrhages during periods of high workload. These delays can negatively impact patient outcomes, increase treatment costs, and reduce overall emergency department efficiency.

***Key Takeaway:** The core problem is not detecting brain hemorrhage—it is prioritizing the right patient at the right time.*

Target User / Customer

The primary users of NeuroTriage AI are **radiologists** working in emergency departments who must rapidly identify critical cases from a large number of incoming CT studies.

Secondary stakeholders include:

- **Emergency physicians**, who depend on timely radiology reports to initiate life-saving treatment.
 - **Hospitals and diagnostic imaging centers**, which aim to improve patient outcomes, operational efficiency, and resource utilization.
 - **Patients**, who benefit from earlier diagnosis and faster clinical intervention.
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Product Concept

NeuroTriage AI is an AI-assisted Emergency Stroke Triage Platform that analyzes non-contrast CT brain studies immediately after image acquisition and estimates the probability of intracranial hemorrhage.

Rather than replacing radiologists or providing autonomous diagnoses, the platform intelligently prioritizes high-risk CT studies within the radiologist's reporting worklist

using AI-generated confidence scores. Cases with high confidence are flagged for immediate review, while lower-confidence studies remain in the normal reporting queue. Every AI recommendation can be accepted, overridden, or rejected by the radiologist, ensuring that clinical responsibility always remains with the healthcare professional.

The proposed **Minimum Viable Product (MVP)** focuses on a single workflow:

Automatically prioritize CT brain studies with suspected intracranial hemorrhage to reduce time-to-diagnosis.

Restricting the MVP to one workflow minimizes implementation complexity while enabling rapid validation of clinical usefulness, workflow integration, and user acceptance.

***Key Takeaway:** NeuroTriage AI improves emergency radiology workflow by prioritizing critical CT studies rather than automating diagnosis.*

Why AI?

Artificial Intelligence is the appropriate solution because this problem requires interpreting complex visual patterns across multiple CT image slices, where deterministic software rules are insufficient. The appearance, size, shape, and location of intracranial hemorrhage vary considerably across patients and cannot be reliably identified using handcrafted rules.

AI image classification models can learn these imaging patterns from annotated historical CT studies and estimate the probability of hemorrhage for every incoming scan within seconds. Rather than replacing clinical expertise, AI functions as an intelligent prioritization engine that enables radiologists to review the most urgent cases first while maintaining complete clinical control through a **Human-in-the-Loop** workflow.

This approach improves workflow efficiency, reduces time-to-diagnosis, and builds clinician trust without compromising patient safety.

***Key Takeaway:** AI is the right lever because the challenge involves probabilistic image interpretation and workflow prioritization, which traditional software cannot perform effectively.*

Recommendation

Recommendation: Pilot

NeuroTriage AI addresses a high-impact clinical workflow where early diagnosis directly influences patient outcomes. The proposed solution demonstrates strong potential to improve emergency radiology efficiency while maintaining clinician oversight. However, because the product operates in a safety-critical healthcare environment, it should not be deployed directly at scale.

I recommend developing a functional MVP followed by a controlled pilot deployment within a single emergency radiology department. During the pilot, both technical AI metrics (Sensitivity, Specificity, False Negative Rate, and Confidence Calibration) and product metrics (Reduction in Time-to-Review, Reduction in Time-to-Treatment, Radiologist Acceptance Rate, Human Override Frequency, and User Satisfaction) should be continuously monitored.

If predefined clinical, operational, and user adoption targets are achieved, the platform can then be progressively scaled across additional hospitals and later extended to detect other neurological emergencies.

Key Takeaway: *A controlled pilot provides the safest and most effective path to validate clinical value, build user trust, and collect evidence before large-scale deployment.*